

● MEDIUM

● INFORMATION AND IDEAS

● ANSWER KEY

Information and Ideas

14

10 answers · Rationale-rich · Teach from doc

Q1**B**

B. They are no longer living species.

Choice B is the best answer because it presents a statement about the two previously unknown penguin species that is supported by the text. According to the text, Alan Tennyson and colleagues studied fossil bones belonging to two previously unknown penguin species that are described as prehistoric, or ancient. The text goes on to say that one of the two species was considerably larger than the emperor penguin, which is currently the largest penguin in existence. Thus, the text indicates that the other species aren't in existence today, or that the two previously unknown penguin species are no longer living species.

Choice A is incorrect. Although the text states that Alan Tennyson is a paleontologist who studied the fossils of two previously unknown penguin species, it doesn't say whether these penguins are frequently studied by paleontologists. Choice C is incorrect because the text says that one of the two previously unknown penguin species was three times larger than the emperor penguin (a penguin alive today), not that they were both smaller than penguin species that exist today. Choice D is incorrect because the text doesn't discuss whether the two previously unknown penguin species spent any time in water.

Q2

B

- B. The average emotionality score of a movie's reviews was a positive predictor of that movie's box office earnings.
-

Choice B is the best answer. This choice weakens the student's conclusion by suggesting that the emotionality of movie reviews is related to box office success: the higher the emotionality score, the better the movie performs at the box office.

Choice A is incorrect. While this choice does mention emotionality scores, it only connects them to star ratings, not to box office success. Choice C is incorrect. The fact that many movies received an average star rating doesn't tell us anything about the relationship between emotionality and box office success. Choice D is incorrect. While this choice suggests that star ratings can predict box office success, it doesn't address the issue of emotionality in written reviews, which is the focus of the student's conclusion.

Q3

A

- A. It isn't growing at the same rate it used to.
-

Choice A is the best answer because it presents an explanation that is directly stated in the text for why ecologists are worried about Pando. The text states that Pando is a colony of about 47,000 quaking aspen trees that represents one of the largest organisms on Earth. According to the text, ecologists are worried that Pando's growth is declining, partly because animals are feeding on the trees. In other words, the ecologists are worried that Pando isn't growing at the same rate it used to.

Choice B is incorrect. Rather than indicating that Pando isn't producing young trees anymore, the text reveals that Pando is indeed producing young trees, stating that those trees can be protected from grazing deer by strong fences. Choice C is incorrect because the text states that fences can be used to prevent deer from eating Pando's young trees, not that Pando itself can't grow in new areas because it's blocked by fences. Choice D is incorrect because the text offers no evidence that Pando's root system is incapable of supporting new trees or is otherwise a cause of worry for ecologists.

Q4

D

D. management, business, and financial services occupations.

Choice D is the best answer because it most effectively uses data from the table to complete the text. The text discusses self-employed workers in the US, and the table shows the US incorporated and unincorporated self-employment rates in four occupational fields in 2015. According to the table, the incorporated self-employment rate was 8.9% for the management, business, and financial services occupational field, which is higher than the incorporated self-employment rate shown for any other occupational field.

Choice A is incorrect because according to the table, the incorporated self-employment rate was 5.8% for the sales and related occupational field, which is lower than the 8.9% for the management, business, and financial services occupational field. Choice B is incorrect because according to the table, the incorporated self-employment rate was 2.7% for the installation, maintenance, and repair occupational field, which is lower than the 8.9% for the management, business, and financial services occupational field. Choice C is incorrect because according to the table, the incorporated self-employment rate was 4.4% for the construction and resource extraction occupational field, which is lower than the 8.9% for the management, business, and financial services occupational field.

Q5

A

A. The slimy threads that hagfish release might help researchers create a new kind of fabric.

Choice A is the best answer. The text first describes hagfish slime and its properties, then it transitions to talking about the possibilities of using lab-made equivalents to use in eco-friendly fabrics.

Choice B is incorrect. The text says the opposite of this choice. The researchers are developing an alternative to petroleum-based fabric. Choice C is incorrect. This choice is too narrow to be the main point of the text. Only one line describes how farming would be “potentially harmful” to the hagfish. Choice D is incorrect. This choice isn’t supported by the text. We don’t know from the text whether being jawless makes the hagfish more vulnerable.

Q6

D

- D. were higher for all three measures, indicating greater overall well-being for these participants.
-

Choice D is the best answer because it most effectively uses data from the table to complete the statement comparing the mean ratings for two different groups of participants in a study. The text explains that a research team evaluated the study's participants using various measures in order to learn whether there might be a medical benefit to telling patients they're receiving a placebo. The table shows the team's mean ratings for participants after 21 days for three of the measures: global improvement, symptom severity reduction, and quality of life improvement. According to the table, the mean ratings were higher for all three measures for participants aware of taking a placebo than for participants in the control group. Given that higher ratings indicate greater well-being, as the text states, the mean ratings in the table indicate greater overall well-being for participants aware of taking a placebo than for participants in the control group.

Choice A is incorrect because the table doesn't include data about individual participants; rather, it presents means, or mathematical averages, of ratings. For this reason, no conclusions can be drawn from data in the table about the extent to which well-being may have varied from participant to participant. Choice B is incorrect because according to the table, the mean ratings for participants aware of taking a placebo were higher for all three measures than for participants in the control group, not lower for two of the measures. Choice C is incorrect because it cites data from the table related to participants in the control group, not to participants aware of taking a placebo. Additionally, the mean ratings in the table for participants aware of taking a placebo are higher for all three measures than for participants in the control group. Given that higher ratings indicate greater well-being, as the text states, the ratings in the table for participants aware of taking a placebo indicate greater well-being for these participants in all three measures.

Q7

B

- B. It has features suggesting that it once held an ocean that underwent gradual sea-level changes over an extended time.
-

Choice B is the best answer. This choice provides direct support for the researchers' conclusion. If the basin once held an ocean of liquid water for "an extended time," that supports the "wet and warm scenario" on Mars, which involved a "prolonged" period of temperate conditions (and not just a "brief period of melting").

Choice A is incorrect. This choice doesn't provide support for the researchers' conclusions. The dimensions of the basin have little bearing on whether or not it was filled with liquid water, and for how long, and that's the evidence that would support the "wet and warm" theory. Similarly, whether or not the basin was formed by an asteroid or a comet is irrelevant to the question of whether or not there was water in the basin for a long period. Choice C is incorrect. This choice does not provide support for the researchers' conclusions, but rather the opposite. A "massive but short lived influx" of liquid water is not the same as a "prolonged phase of warm, wet conditions." It more reflects the opposing theory, a "brief period of melting." Choice D is incorrect. This choice does not support Cardenas and Lamb's conclusion. Both theories about liquid water on Mars ("wet and warm" and "brief period of melting") involve flowing water, but lava isn't mentioned at all in the text. Therefore, this choice doesn't provide exclusive support for either theory.

Q8

B

- B. at 48 hours, more than 70% of seeds treated with H₂S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 50% of untreated seeds had germinated.
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Choice B is the best answer. The claim is that some concentrations of H₂S led to increased germination rates, and this choice accurately shows that seeds treated with 10 micromoles per liter of H₂S tended to germinate faster than untreated seeds.

Choice A is incorrect. This choice doesn't justify the claim. The claim compares the germination rates of seeds exposed to certain concentrations of H₂S to untreated seeds, but this choice only discusses one concentration of H₂S, so it can't support any comparison between treated and untreated groups. Choice C is incorrect. This choice misreads the graph. At 168 hours, only about 85% of seeds treated with H₂S at 500 micromoles per liter and well over 70% of untreated seeds had germinated (about 80%). Choice D is incorrect. This choice misreads the graph. At 48 hours, about 70% of seeds treated with H₂S at 10 micromoles per liter and about 50% of untreated seeds had germinated.

Q9

D

- D. Species that build open nests tend to have larger ranges than species that build domed nests.
-

Choice D is the best answer because it presents a finding that, if true, would challenge the assumption that many ecologists have made about the connection between the building of domed nests and geographic range in songbirds. The text says that many ecologists have assumed that since domed nests provide protection from weather conditions, songbird species that build such nests should be able to have larger geographic ranges than songbird species that build open nests do. If Medina and her colleagues found that species that build open nests tend to have larger geographic ranges than species that build domed nests do, their finding would show the opposite of what the ecologists have assumed. It would therefore challenge the ecologists' assumption.

Choice A is incorrect because nothing in the text suggests that there's a relationship between songbird species' extinction rates and their geographic ranges. The finding that species that build open nests tend to have higher extinction rates than species that build domed nests do would therefore have no clear bearing on the ecologists' assumption that domed nests allow species that build them to have larger geographic ranges than those of species that build open nests. Choice B is incorrect because nothing in the text suggests that there's a relationship between songbird species' sizes and their geographic ranges. The finding that species that build open nests tend to be smaller in size than species that build domed nests are would therefore have no clear bearing on the ecologists' assumption that domed nests allow species that build them to have larger geographic ranges than those of species that build open nests. Choice C is incorrect because although the text indicates that many ecologists have assumed that there's a connection between how songbird species build their nests and the species' geographic ranges, the text says that this assumption is based on the shape of the nests—that is, whether the nests are domed or open—not the number of materials used. The finding that species that build open nests tend to use fewer materials to build their nests than species that build domed nests do would therefore have no clear bearing on the ecologists' assumption that domed nests allow species that build them to have larger geographic ranges than those of species that build open nests.

Q10

C

- C. fewer than 100 suggestions were coded as subtractive, whereas more than 550 suggestions were coded as additive.
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Choice C is the best answer. This choice shows that people suggested removing things to achieve improvements a lot less often than they suggested adding things, which supports the claim that people tend not to think of removing things as a likely way to improve the university.

Choice A is incorrect. This choice doesn't support the claim. We are looking to prove that people suggested subtractive means of achieving improvements less often than other means, but this choice doesn't address how often people made subtractive suggestions. Choice B is incorrect. This choice doesn't support the claim. Invalid responses were incomprehensible, so we can't compare what they were suggesting to what was being suggested in subtractive responses. Choice D is incorrect. This choice misreads the graph. Fewer than 100 suggestions were coded as subtractive. 175 is the approximate number of suggestions coded as "neither additive or subtractive."

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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